H-B-1049
August 21, 1974
SUPERSEDING
Fed. Spec. H-B-173 (In Part)
October 8, 1948

FEDERAL SPECIFICATION

BRUSH, FLASK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a brush, primarily intended for use in cleaning laboratory flasks.

1.2 Classification. The flask brush covered by this specification will be one type, class, and size.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-S-951 - Synthetic Filaments for Brushes, General Specification for.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specification:

MIL-W-6110 - Wood, Determination of Moisture Content of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Wood. Wood for the handle shall be close-grained and straight-grained hardwood, such as birch, beach, maple, or ash. The wood shall be thoroughly seasoned so that when tested in accordance with 4.2.2, it shall have a moisture content of not more than 12 percent.

3.1.2 Synthetic filament. The filament for the brush head shall be made of synthetic polymeric amide (generic term: nylon), meeting the requirements of class N, type II or III in H-S-951. The color of the filament shall be black.

3.1.3 Wire. The wire shall be made from a metal which is inherently corrosion resistant or a steel which has been coated to resist corrosion.

3.1.4 Plastic. Plastic for handles shall be of any type which will not soften or be materially affected by hot water or cleaning agents.

3.2 Construction.

3.2.1 Handle. The handle shall be made of wood (see 3.1.1) or plastic (see 3.1.4) conforming to the requirements of Table I. The handle shall be either rounded or tapered, or round and straight beyond the brush part. The portion within the brush part shall be spirally grooved to aid in retaining the filaments.

3.2.1.1 Handle end. Provisions shall be made for hanging the brush by the handle, when not in use, by either of the following methods:

A 3/16 inch, plus or minus 1/16 inch, diameter hole located 5/8 inch, plus or minus 1/8 inch, from the end.
A corrosion-proof metal hook attached to the handle end.

3.2.2 Brush head. The synthetic filaments shall meet the requirements of 3.1.2 and Table I. The filaments shall be held at their centers in spiral grooves of the handle by tightly-wound wire conforming to the requirements of 3.1.3 and Table I. The wire ends shall be turned in and securely tacked.

3.3 Physical requirements. The flask brush shall conform to the physical requirements in Table I.

3.4 Workmanship. The finished brush shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's), and they shall be free from defects which affect their appearance or which may affect their serviceability.

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TABLE I. Physical requirements of flask brushes

Characteristics	Dimension
Length, overall (inches)	15 ± 1/2
Brush part:	
Length (including dome end) (inches)	5-1/2 ± 1/4
Diameter (inches)	3 ± 1/4
Filament:	
Length (minimum inches) (projection from handle to extremity of filament)	1-1/8
Diameter (inch)	.0125 ± .0005
Depth of spiral grooves (inch)	5/32 ± 1/32
Handle	
Length from base of brush part (inches)	9-1/2 ± 1/2
Diameter (inch)	3/4 ± 1/16
Wire:	
Diameter (minimum inch)	0.028

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2.2 In-process inspection.

4.2.2.1 Moisture content of handles (wood). Prior to being machined and finished, the handles shall be tested for moisture content in accordance with MIL-W-6110 and then examined for compliance with 3.1.1. Three determinations shall be taken on each sample unit and the results shall be the average of the three. The lot shall be the number of handles being manufactured under essentially the same conditions and to be offered for inspection at one time. The inspection level shall be S-1, with an AQL of 6.5 expressed in terms of defects per hundred units.

4.2.3 End item inspection. Examination of the end item shall be in accordance with the classification of defects, inspection levels, and AQL's set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of brushes for examination under 4.2.3.1 and 4.2.3.2, in units of intermediate containers for examination under 4.2.3.3, and in units of shipping containers for examination under 4.2.3.4.

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4.2.3.1 Examination of the end item for defects in appearance, construction, and workmanship. The sample unit for this examination shall be one brush.

Examine	Defect
Material	Bristle not synthetic filament specified. Wire not made of corrosion-resistant metal or steel coated to resist corrosion. Wood handle not made of a close, straight-grained hardwood.
Construction and appearance (filament)	Not uniformly and firmly held in place; can be removed by hand. Not evenly distributed in brush head; not uniform in length. Synthetic filament not black in color and not diameter specified.
Brush head	Not provided with a tuft extending beyond the end of brush as specified. Tuft not securely bound with wire. Filament not approximately parallel to each other, not at an angle approximately 90 degrees to the handle, and not evenly distributed in the brush head.
Handle	Not specified shape; hand end not provided with specified means for hanging.
Workmanship	Filament not neatly and evenly trimmed; not trimmed as specified. Wood handle not smoothly finished in accordance with commercial practice. Sharp edge or point on wire; plating rough, thin, or has bare spots.

4.2.3.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one brush.

Examine	Defect
Dimensions	Any dimension not in accordance with specification requirements.

4.2.3.3 Examination of the end item for count. The sample unit for this examination shall be one intermediate container. The average count per container shall be not less than the specified or indicated quantity.

4.2.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking requirements, as specified, are complied with. The sample unit for this examination shall be one shipping container fully prepared for delivery.

Examine	Defect
Packaging	Not packaged as specified; intermediate container not as specified.
Packing	Not type shipping container specified. Any break, tear, hole, or other defect in container which might cause damage to or loss of content. Waterproof case liner, when applicable, missing or not as specified.
Content	More or less than quantity of units or intermediate containers indicated or specified.
Marking	Omitted, incomplete, incorrect, illegible; not in accordance with contract requirements.

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4.2.3.5 Inspection levels and AQL's for examinations.

Examination paragraph	Inspection level	AQL
4.2.3.1	I	4.0
4.2.3.2	S-1	2.5
4.2.3.3	S-2	---
4.2.3.4	S-2	4.0

4.3 Testing of end item. The end item shall be tested for the applicable characteristics as specified in 4.4. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.4 Tests.

4.4.1 Synthetic filaments. The synthetic filaments shall be tested as described in H-S-951.

4.4.1.1 The filament shall be tested for compliance with 3.1.2.

4.4.2 Diameter of wire. The diameter of the wire taken from the brushes shall be determined by means of a micrometer caliper graduated to read 0.001 inch or less. Any uneveness in the wire shall be removed before measuring with the micrometer. The wire shall meet the requirements of 3.2.3.

4.4.3 Length of filament. The length of the filament shall be determined by means of a scale or tape graduated in 1/16 inch or less.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A or B. Twelve brushes shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676. The brushes shall be arranged reversed end to handle order within the box. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level C. The brushes shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Six packages (72 brushes) shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be waterproof sealed with tape and reinforced with strapping in accordance with the appendix to the box specification.

5.2.2 Level B. The brushes shall be packed as specified in 5.2.1, except class domestic boxes shall be utilized. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The brushes shall be packed to afford adequate protection against physical and environmental damage during shipment, handling, and limited storage. The manufacturer's commercial practice is acceptable when it meets these requirements.

5.3 Marking. In addition to any marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

6. NOTES

6.1 Intended use. This brush is primarily intended for use in cleaning laboratory flasks.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

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(a) Title, number, and date of this specification.
(b) Responsibility for inspection (see 6.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Comparison. Comparison of classification between Federal Specification H-B-173 and H-B-1049 is as follows:

Brush type VII of H-B-173 is now described in H-B-1049.

6.4 Supersession. That part of Federal Specification H-B-173 that pertains to type VII flask brushes is superseded by this Federal specification. The remainder of H-B-173 will remain in force until Federal Specifications H-B-1050A and H-B-1051A are published with appropriate supersession data, at which time a cancellation will be published for H-B-173.

MILITARY INTEREST

DOD has waived coordination

PREPARING ACTIVITY: GSA-FSS

User Interest:

Army-GL

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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